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Illustration 10.15 C++ program to sum numbers from 1 to 10 using while loop

```
#include <iostream>
using namespace std;
int main ()
{
int i=1,num,avg,sum=0;
while(i<=5)
{
        cout<<"Enter the number : ";
        cin>>num;
        sum=sum+num;
        i++;
}
avg=sum/5;
cout<<"The sum is "<<sum<<endl;
cout<<"The average is "<<avg;
return 0;
}
```

Output

```
Enter the number : 1
Enter the number : 2
Enter the number : 3
Enter the number : 4
Enter the number : 5
The sum is 15
The average is 3
```

In the above program, integer variables **num** and **avg** are declared and variable i is initialized to 1 and sum to 0. The while loop checks the condition, since i <= 5 the condition is true, a number is read from the user and this is added to sum and i is incremented by 1. Now, the condition is i <= 5 is again checked. Since 2 <=5, the second number is obtained from the user and it is added to sum. This continues, until i becomes 6, at which point the while loop terminates. After the loop termination, the avg is computed and both sum and avg are displayed.

While loop variation

A while loop may contain several variations. It can be an empty loop or an infinite loop. An empty while loop does not have any statement inside the body of the loop except null statement i.e., just a semicolon.